COE Applied AI Workshop

Monday July 13

Time	Activity	Lead
1:00 pm - 2:15 pm	Introduction to AI	Edgar Lobaton
2:30 pm - 3:00 pm	Hands on: Exploring Code Assistants	Edgar Lobaton
3:00 pm - 3:30 pm	Presentation: Playing with AI - Curiosity, Mistakes, and Finding Real Co-Intelligence	Tarek Aziz
	Tarek Aziz is an Associate Professor in Civil, Construction, and Environmental Engineering. He is an environmental process engineer interested in nature-based solutions for water treatment and water quality. His work involves modeling, experimentation, and field work and in the past years, he's woven AI into all three. In his spare time Tarek loves spending time with family, traveling, reading and playing video games.	
3:00 pm - 3:45 pm	Coffee Break	
3:45 pm - 4:15 pm	Tool Overview: GitHub	Tarek Aziz
4:15 pm - 5:00 pm	Software Setup Session	Edgar Lobaton

Tuesday July 14

Time	Activity	Lead
1:00 pm - 2:00 pm	AI Concepts	Edgar Lobaton
2:00 pm - 2:30 pm	Reflection: Exploring Code Assistants	Edgar Lobaton
2:30 pm - 3:00 pm	Presentation: Towards open autonomous labs for plant and microbial agentic phenotyping	Larry York, ORNL
	Larry York is a whole-plant and soil ecophysiologist at Oak Ridge National Laboratory. He develops theory, hardware, and software to increase understanding of how roots interact with microbes in soil. He has developed tools such as RhizoVision Explorer for measuring root system architectural traits and high-throughput respiration and nitrogen uptake assays. Over the past couple years, this co-development of hardware and software has led him to explore the use of robotics and large language models (LLMs) to enable autonomous phenotyping. This talk will focus on open-source robotic phenotyping and new data analysis benchmarks using local LLMs.	
3:00 pm - 3:15 pm	Coffee Break	
3:15 pm - 3:45 pm	Tool Overview: Virtualization	Edgar Lobaton

3:45 pm - 4:15 pm	Presentation: Pedagogical Perspective	Edgar Lobaton
4:15 pm - 4:45 pm	Hands on: GitHub	Tarek Aziz
4:45 pm - 5:00 pm	Open Discussion | Survey	Edgar Lobaton